Jennifer Marroquin

954-419-6202 | jenmarroquin11@gmail.com | *https://www.linkedin.com/in/jennifermarroquin11/*

Education

University of Central Florida *Expected Graduation:* ***December 2028***
Bachelor of Science in Electrical Engineering *Orlando, FL*

Relevant Coursework: Calculus I – III, Physics I – II, Engineering Analysis & Computation, Digital Systems

Honors & Awards: 1st Place IEEE UCF Competition, Bright Future Medallion Recipient

Skills

Programming & Tools: C++, MATLAB, Verilog, Arduino, MSP430, Bluetooth Modules, Microsoft Office

Hardware & Embedded Systems: Motor Control, Sensor Interfacing, Circuit Design & Validation, Embedded Firmware Development, Data Management, FPGA Simulation

Professional Skills: Team Collaboration, Analytical Problem-Solving, Adaptability, Time Management

Languages: English (Fluent), Spanish (Native)

Projects

Custom Embedded RC Derby Car | *ESP32, Control Systems, Power Electronics* *January 2026 – Present*

- Leading end-to-end electrical system architecture for a 3-minute competitive RC racing platform, including power distribution, motor control, and embedded firmware integration.
- Designing and validating a regulated multi-rail LiPo (7.4V) power architecture to maintain stable ESP32 operation under transient current spikes and dynamic load conditions.

Digital Logic Design & Verification | *Verilog, FPGA Simulation* *January 2026 – Present*

- Designing and optimizing combinational logic circuits using Boolean algebra and Karnaugh map minimization to reduce gate count and propagation delay.
- Developing comprehensive testbenches to verify functional correctness across exhaustive and edge-case input permutations.
- Simulating and analyzing timing behavior in Vivado, validating logic transitions, propagation characteristics, and equivalence between SOP and POS representations.

Remote-Joystick Controlled Car | *Arduino, Bluetooth, Motors December 2024 – April 2025*

- Designed and implemented a remote-controlled car with Arduino microcontroller and Bluetooth communication.
- Integrated circuits, motor drivers, and UI programming for IEEE showcase demos.
- Awarded 1st place in IEEE UCF Project Competition.

TI-RSLK Robot Maze | *C/C++, Embedded Systems, Sensors September 2024 – November 2024*

- Programmed a TI-RSLK robot in C/C++ to autonomously navigate a maze using real-time sensor feedback, achieving 100% maze completion across multiple test runs.
- Implemented path-planning algorithms and integrated multiple 3+ hardware modules.
- Debugged real-time system performance in a lab environment, achieving reliable maze completion.

Great Navel Orange Race Autonomous Boat | *MSP430, MATLAB, Motor Control Jan 2025 – April 2025*

- Collaborated with a team of 4 engineering students to design and program an autonomous boat for UCF's GNOR
- competition.
- Programmed MSP430 microcontroller for motor control and heading-angle navigation.
- Simulated obstacle avoidance in MATLAB and optimized pathing for 8-minute timed pond traversal.

Extracurriculars

Society of Hispanic Professional Engineers – SHPE September 2025 – Present
Society of Women Engineers – SWE August 2024 – Present
Institute of Electrical and Electronics Engineers – IEEE October 2024 – Present
Women in Electrical Engineering and Computer Science – WEECS August 2024 – Present

Work Experience

Server – Bar Louie *January 2025 – Present*

- Balancing a 20–30-hour per week schedule alongside a full-time course load.
- Delivered high-quality customer service while managing 5-8 tables in a fast-paced environment.
- Resolved issues under pressure and collaborated with team members to maintain smooth operations.